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Inventor(s)

Kita; Keiji et al.

Mandarin plant named 'Mikanken4'

Abstract

A new variety of citrus 'Mikanken4' that has dark orange fruit color, without seeds, high juiciness, orange-like flavor, ease of peeling the fruit by hand, good appearance, and good tasting qualities.

Latin Name: Citrus spp. Mikanken4

Inventors: Kita; Keiji (Ehime, JP), Nakata; Haruhito (Ehime, JP), Shigematsu; Yukinori (Ehime, JP), Hyodo; Yoji (Ehime, JP), Mido; Hiroaki (Ehime, JP), Kikuchi; Takehiro (Ehime, JP), Masamoto; Yasuyuki (Ehime, JP), Sakimoto; Takae (Ehime, JP), Yakushiji; Hiromichi (Ehime, JP), Okusada; Takehiro (Ehime, JP), Ninomiya; Taizo (Ehime, JP), Okamoto; Mitsutoshi (Ehime, JP), Yamamoto; Saki (Ehime, JP), Miyoshi; Saki (Ehime, JP), Hioki; Toya (Ehime, JP), Kinoshita; Kei (Ehime, JP), Sugawara; Daijiro (Ehime, JP)

Applicant: EHIME PREFECTURE (Ehime, JP)

Assignee: EHIME PREFECTURE (Matsuyama, JP)

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Background/Summary

- (1) Common name: Mandarin.
- (2) Botanical classification: *Citrus* spp.
- (3) Variety denomination: 'Mikanken4'.

BACKGROUND OF THE NEW VARIETY

(4) The present invention relates to a new and distinct variety of Mandarin, *Citrus* L., which has been given the variety denomination 'Mikanken4'.

(5) ORIGIN OF THE VARIETY

- (6) The parent varieties are 'Kiyomi' (female parent) (not patented in the U.S.) and MINEOLA' (male parent) (not patented in the U.S.), respectively.
- (7) The variety was developed and propagated in Uwajima, Ehime, Japan.
- (8) The variety was reproduced asexually by vegetative reproduction by top grafting one-year-old scions on rootstocks in Uwajima, Ehime, Japan. The rootstock variety was 'MIYAGAWAWASE' (not patented in the U.S.)
- (9) In 1989, the parent varieties were crossed.
- (10) In 1990, collected seeds were grown.
- (11) In 1992, saplings were grafted on top of rootstock 'MIYAGAWAWASE' (not patented in the U.S.)
- (12) In 1995, the first fruit was harvested, and lines were primary selected.
- (13) In 1997, plants were secondary selected, and the variety was named 'Mikanken4'.
- (14) In 2024, breeding was completed with determination of distinct, uniform, and stable (DUS) status confirmed, and the variety was named 'Mikanken 4'.
- (15) Cultivation of the variety does not require special conditions or treatments.
- (16) The variety is currently mainly intended for fruit production.

(17) SUMMARY OF THE VARIETY

- (18) The new mandarin plant 'Mikanken4' has leaf blades with very small length/width ratio, fruit with dark orange surface color, and late maturity for consumption (typically May), which distinguish 'Mikanken4' from its female parent 'Kiyomi'.
- (19) Further, the new mandarin plant 'Mikanken4' has fruit with an absence of neck, which distinguishes it notably from its male parent 'MINEOLA'.
- (20) Also, the new mandarin plant 'Mikanken4' having leaf blades with very small length/width ratio, and late maturity for consumption (typically May), and its fruit having a proximal part with a flattened general shape, distinguish it notably from the comparative variety 'Ehimekashi48',
- (21) A comparison with the female parent 'KIYOMI' is as follows (evaluation based on averages):
- (22) TABLE-US-00001 TABLE 1 Female parent Present variety No. Characteristics 'KIYOMI' 'Mikanken4' 16 Leaf blade: ratio length/width Small (2.2) Very small (1.9) 83 Fruit surface: predominant Yellow orange Orange color(s) RHS 17A RHS N25C 119 Time of maturity of fruit for Medium (March) Late (May) consumption
- (23) A comparison with the male parent 'MINEOLA' is as follows (evaluation based on averages):
- (24) TABLE-US-00002 TABLE 2 Male parent Present variety No. Characteristics 'MINEOLA' 'Mikanken4' 66 Fruit: presence of neck Present Absent
- (25) A comparison with the reference variety 'Ehimekashi48' is as follows (evaluation based on averages):
- (26) TABLE-US-00003 TABLE 3 Reference variety Present variety No. Characteristics 'Ehimekashi48' 'Mikanken4' 16 Leaf blade: ratio length/width Small (2.2) Very small (1.9) 65

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the accompanying drawings, which are as nearly true as is reasonable possible to make in a color illustration of this type:
- (2) FIG. 1 is a color photograph showing a general view of the mandarin plant 'Mikanken4' bearing fruits;
- (3) FIG. 2 is a color photograph showing leaves of 'Mikanken4'
- (4) FIG. 3 is a color photograph showing flowers of 'Mikanken4';
- (5) FIG. 4 is a color photograph showing fruits of 'Mikanken4';
- (6) FIG. 5 is a color photograph showing the leaves of 'Mikanken4' and its female parent 'KIYOMI', illustrating the difference in the predominant color of fruit surface;
- (7) FIG. 6 is a color photograph showing fruits of 'Mikanken4' and the comparative variety 'Ehimekashi48', illustrating the difference in the general shape of the proximal part.
- (8) Due to chemical and/or digital development, processing and printing, the plants or portions of plants depicted in the photographs may or may not be precisely accurate, when compared to the actual botanical specimens.

DETAILED DESCRIPTION OF THE INVENTION

- (9) The botanical description and the photographs are of the 'Mikanken4' plants evaluated in 2024 (including for the comparison with the reference plants 'KIYOMI' and 'Ehimekashi48'). Values as provided are averages.
- (10) The plants shown on the photographs were grown at Uwajima, Ehime, Japan. The plant is a 15-year grown after top grafting. The tree was photographed on Mar. 25, 2024, the flowers were photographed on Apr. 25, 2024, the mature leaves were photographed on Mar. 25, 2024, and the fruit were photographed on Mar. 29, 2024.
- (11) Colors are given according to the RHS (Royal Horticultural Society) Colour Chart (Fourth Ed., 2001).

BOTANICAL DESCRIPTION

- (12) TABLE-US-00004 TABLE 4 No. Characteristics States Values 1. Ploidy Diploid 2. Tree: growth habit Spreading 3. Tree: density of spines Intermediate 4. Tree: length of spines Short 2.2 cm 5. Tree height 2.7 m 6. Tree spread 3.2 m 7. Trunk diameter 27.7 cm 8. Trunk texture Rough 9. Trunk color Green brown RHS 152B 10. Crotch angle 47.5° 11. Branch length Short 9.8 cm 12. Branch texture Smooth 13. Branch color Brown green RHS 146A 14. Leaf blade: length (apical leaflet in case Medium 8.0 cm of compound leaf) 15. Leaf blade: width (as for 5) Medium 4.3 cm 16. Leaf blade: ratio length/width (as for 5) Very small 1.9 17. Leaf blade: shape in cross section Intermediate (as for 5) 18. Leaf blade: twisting Absent or weak 19. Leaf blade: blistering Absent or weak 20. Leaf blade: green color Medium 21. Leaf blade: undulation of margin Intermediate 22. Leaf blade: incisions of margin Absent 23. Leaf blade: shape of apex Acute 24. Leaf blade: emargination at tip Present 25. Leaf blade size Medium 22.4 cm.sup.2 (length 8.0 cm, width 4.3 cm) 26. Leaf arrangement Alternate 27. Leaf type Simple leaf 28. Leaf shape Fusiform 29. Leaf color: upper surface Dark green RHS 141A 30. Leaf color: lower surface Brown green RHS 137D 31. Petiole: length Medium 1.8 cm 32. Petiole: presence of wings Present 33. Petiole diameter Medium 1.8 mm 34. Petiole color Dark green RHS 141A 35. Petiole wings Present 36. Petiole attachment (Not examined) 37. Petiole texture Smooth 38. Varieties with petiole wings present Narrow 3.3 mm only: Petiole: width of wings 39. Flower: type Single 40. Flower: diameter of calyx — 0.9 cm 41. Flower: number of petals 5 42. Flower: sepal size length 0.55 cm, width 0.93

cm 43. Flower: sepal shape Synsepalous calyx 44. Flower: sepal color Light green RHS 144D 45. Flower: length of petal Short 13.7 mm 46. Flower: width of petal Medium 6.0 mm 47. Flower: ratio length/ width of petal — 2.3 48. Petal color: upper surface White RHS 155A 49. Petal color: lower surface White RHS 155A 50. Petal texture Smooth 51. Flower: length of stamens — 9.2 mm 52. Flower: blooming period Medium May 1-10 53. Flower: fragrance Present 54. Flower: fertility Present 55. Pollen abundance Absent 56. Pollen color (Not examined) 57. Anther: color Light yellow RHS 8D 58. Style: length — 6.7 mm 59. Inflorescence: clustering of fruits Present 60. Fruit: length — 7.4 cm 61. Fruit: diameter — 8.6 cm 62. Fruit: ratio length/diameter Medium 1.16 63. Fruit: position of broadest part At middle 64. Fruit: shape in transverse section Circular 65. Fruit: general shape of proximal part Flattened (excluding neck, collar and depression at stalk end) 66. Fruit: presence of neck Absent 67. Fruit: presence of depression at stalk Present end 68. Fruit: depth of depression at stalk end — 2.0 mm 69. Fruit: presence of constriction at stalk Absent end 70. Fruit: number of radial grooves at Intermediate stalk end 71. Fruit: length of radial grooves at stalk Short to 10.1 end medium 72. Fruit: presence of collar Absent 73. Fruit: abscission layer between floral Absent or disc and fruit weakly developed 74. Fruit: general shape of distal part Flattened (excluding nipple, bulging of navel and depression at distal end) 75. Fruit: presence of depression at distal Present end 76. Fruit: depth of depression at distal end — 2.6 mm 77. Fruit: diameter of depression at distal — 14.5 mm end 78. Fruit: presence of areola Absent 79. Fruit: diameter of stylar scar — 2.1 mm 80. Fruit: persistence of style None 81. Fruit: presence of navel opening Absent 82. Fruit: presence of radial grooves at Absent distal end 83. Fruit surface: predominant color(s) Orange RHS N25C 84. Fruit surface: glossiness Medium 85. Fruit surface: roughness Smooth 86. Fruit surface: size of oil glands All more or less the same size 87. Fruit surface: size of larger oil glands Small 88. Fruit surface: conspicuousness of larger Strong oil glands 89. Fruit surface: presence of pitting and Pitting and pebbling on oil glands pebbling absent 90. Fruit rind: thickness Medium 3.5 mm 91. Fruit weight Heavy 286 g 92. Fruit rind: adherence to flesh Weak 93. Fruit rind: strength Weak 94. Fruit rind: oiliness Dry 95. Fruit rind: conspicuousness of oil Absent or glands on inner surface weakly conspicuous 96. Fruit: color of albedo Light yellow RHS 8C 97. Fruit: density of albedo Medium 98. Fruit: amount of albedo adhering to Small flesh (strands excluded) 99. Fruit: presence of albedo strands Absent 100. Fruit: main color of flesh Dark orange RHS N25C 101. Fruit: filling of core Absent or very sparse 102. Fruit: diameter of core Very small 10.9 mm 103. Fruit: presence of rudimentary Absent or segments very weak 104. Fruit: number of well-developed — 12 segments 105. Fruit: coherence of adjacent segment Strong walls 106. Fruit: strength of segment walls Weak 107. Fruit: length of juice vesicle — 6.2 mm 108. Fruit: thickness of juice vesicles — 2.6 mm 109. Fruit; conspicuousness of juice High vesicle walls 110. Fruit: coherence of juice vesicles Medium 111. Fruit: presence of navel (viewed Absent or internally) very rare 112. Fruit: juiciness High 113. Fruit juice: total soluble solids High 12.8 (Brix) 114. Fruit juice: acidity High 1.95 mg/100 ml 115. Fruit: strength of fiber Weak 116. Fruit: number of seeds (controlled Absent or 0 manual self-pollination) very few 117. Fruit: number of seeds (open Absent or 0 pollination) very few 118. Seed: polyembryony Absent 119. Time of maturity of fruit for Late May consumption 120. Productivity (pounds or bushels per (Not acre) examined) 121. Fruit: parthenocarpy Present

(13) Other features of the plant are as follows: Disease and Pest resistance: Normal resistance was observed for pests and diseases in Uwajima, Ehime, Japan. Tree hardiness: USDA Hardiness Zone Pa-9b. Fruit ship ability: medium, slightly inferior compared to the variety ‘IYOKAN’ which has excellent storage ability. Fruit storage ability: medium.

Claims

1. A new and distinct variety of citrus named ‘Mikanken4’, substantially as described and

illustrated herein.
